

BOTANY HUB MULTAI

B.Sc. I BOTANY — MINOR PAPER II

DIVERSITY OF PLANTS

UNIT I — Plants in Indian Knowledge Systems

DETAILED TEACHER-STYLE NOTES

Definitions • Complete explanations • Classification tables • Life cycles • Flowcharts • Diagrams • Examples • Economic importance • Exam points

Prepared strictly from the uploaded 2025–26 Minor Paper II syllabus.

UNIT I — PLANTS IN INDIAN KNOWLEDGE SYSTEMS

The syllabus specifically requires the contributions of M.O.P. Iyengar, Shiv Ram Kashyap, S.S. Bir, Birbal Sahni and D.D. Pant, plus names and descriptions of plant groups in ancient Indian texts. ■filecite■turn11file0■L35-L46■

1. Plant diversity and Indian knowledge systems

Definition — Indian knowledge systems: Bodies of knowledge developed and transmitted in the Indian subcontinent through texts, observation, teaching and practical traditions, including knowledge relevant to agriculture, useful plants, medicine and the natural world.

For Botany, the key point is that historical plant knowledge and modern scientific botany should be connected carefully. Traditional descriptions are valuable for history and documentation; modern taxonomy and experimental methods are used to test and standardize biological claims.

From traditional observation to modern botanical study



2. M.O.P. Iyengar — Indian phycology

M.O.P. Iyengar is remembered as a major Indian phycologist whose work helped establish systematic study of algae in India. In an exam answer, explain his contribution in the context of algal diversity, taxonomy and development of phycological research rather than writing only his name.

Scientist	Field	Contribution theme
M.O.P. Iyengar	Phycology	Major contributions to the systematic study and development of algal research in India

3. Shiv Ram Kashyap — Indian bryology

Shiv Ram Kashyap made important contributions to the study of liverworts and bryophytes of the western Himalayas and adjoining regions. His work is important because bryophyte diversity requires careful field collection, description and comparison of structures and reproductive stages.

4. S.S. Bir — Indian pteridology

S.S. Bir is associated with important work on pteridophytes and their diversity in India. Pteridology deals with ferns and related vascular cryptogams, including their morphology, anatomy, taxonomy and distribution.

5. Birbal Sahni and D.D. Pant — gymnosperm and palaeobotanical studies

Birbal Sahni is especially associated with Indian palaeobotany and fossil plant studies. D.D. Pant contributed to gymnosperm and fossil-plant research. Together, these names are important for understanding how fossil evidence helps reconstruct plant evolution and the history of seed plants.

Name	Main association	Why it matters
M.O.P. Iyengar	Algae	Development of Indian phycology
Shiv Ram Kashyap	Bryophytes/liverworts	Documentation and study of bryophyte diversity
S.S. Bir	Pteridophytes	Indian pteridological research
Birbal Sahni	Palaeobotany	Fossil plants and plant evolution
D.D. Pant	Gymnosperm/fossil plant studies	Seed-plant and palaeobotanical research

6. Plant groups in ancient Indian texts

Ancient sources contain names and groupings based on visible form, habitat, use, life habit and cultural significance. These groupings do not always correspond exactly to modern taxonomic ranks. A strong answer should state this difference clearly.

Basis of grouping	Traditional practical question	Modern botanical parallel
Habit	Tree, shrub, herb, climber	Growth form
Use	Food, medicinal, fibre, timber plant	Economic botany
Habitat	Aquatic, terrestrial, forest plants	Ecology
Visible structure	Flowering/non-flowering or other obvious features	Morphological classification

Poster activity

Make a one-page poster: portrait/name → field → 3 contribution points → major plant group studied → one historical significance point.

Exam focus: Unit I long answer: start with the field, then scientist-wise contributions in separate headings, add a comparison table, and end with the importance of preserving historical botanical knowledge. The syllabus explicitly lists these contributors and the ancient-text component. ■filecite■turn11file0■L36-L46■